Topic: Exponential growth and decay

Question: If an amoeba population doubles in 4 hours, and assuming normal exponential growth, find the growth constant for the population.

Answer choices:

A $k \approx 0.17$

B $k \approx 0.50$

C $k \approx 0.69$

D $k \approx 5.77$

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Solution: A

The size of the original population is P_0. Since the population doubles after 4 hours, we know $P(4) = 2P_0$. Plug this initial condition into the exponential growth equation.

$$P(t) = P_0 e^{kt}$$

$$2P_0 = P_0 e^{k(4)}$$

$$2 = e^{4k}$$

Take the natural log of both sides.

$$\ln 2 = \ln(e^{4k})$$

$$\ln 2 = 4k$$

$$k \approx 0.17$$

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Topic: Exponential growth and decay

Question: If a bacteria population grows exponentially and increases fivefold in 6 hours, how long did it take for the population to double?

Answer choices:

A $t \approx 4.16$ hours

B $t \approx 13.93$ hours

C $t \approx 2.58$ hours

D $t \approx 0.43$ hours

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Solution: C

The size of the original population is P_0. Since the population increases fivefold in 6 hours, we know $P(6) = 5P_0$. Plug this initial condition into the exponential growth equation.

$$P(t) = P_0 e^{kt}$$

$$5P_0 = P_0 e^{k(6)}$$

$$5 = e^{6k}$$

Take the natural log of both sides.

$$\ln 5 = \ln(e^{6k})$$

$$\ln 5 = 6k$$

$$k = \frac{\ln 5}{6}$$

This is the growth constant for the population, and we can plug it back in to figure out how long it takes for the population to double (reach a size of $2P_0$).

$$P(t) = P_0 e^{kt}$$

$$2P_0 = P_0 e^{\frac{\ln 5}{6}t}$$

$$2 = e^{\frac{\ln 5}{6}t}$$

Take the natural log of both sides.

$$\ln 2 = \ln(e^{\frac{\ln 5}{6}t})$$

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$$\ln 2 = \frac{\ln 5}{6}t$$

$$t = \frac{6 \ln 2}{\ln 5}$$

$$t \approx 2.58$$

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Topic: Exponential growth and decay

Question: If a fruit fly population grows exponentially and increases sixfold in 7 days, how long did it take for the population to triple?

Answer choices:

A $t \approx 11.42$ days

B $t \approx 3.50$ days

C $t \approx 1.77$ days

D $t \approx 4.29$ days

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Solution: D

The size of the original population is P_0. Since the population increases sixfold in 7 days, we know $P(7) = 6P_0$. Plug this initial condition into the exponential growth equation.

$$P(t) = P_0 e^{kt}$$

$$6P_0 = P_0 e^{k(7)}$$

$$6 = e^{7k}$$

Take the natural log of both sides.

$$\ln 6 = \ln(e^{7k})$$

$$\ln 6 = 7k$$

$$k = \frac{\ln 6}{7}$$

This is the growth constant for the population, and we can plug it back in to figure out how long it takes for the population to triple (reach a size of $3P_0$).

$$P(t) = P_0 e^{kt}$$

$$3P_0 = P_0 e^{\frac{\ln 6}{7}t}$$

$$3 = e^{\frac{\ln 6}{7}t}$$

Take the natural log of both sides.

$$\ln 3 = \ln(e^{\frac{\ln 6}{7}t})$$

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$$\ln 3 = \frac{\ln 6}{7}t$$

$$t = \frac{7 \ln 3}{\ln 6}$$

$$t \approx 4.29$$

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